

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 18****Topics covered :**

1. Boundary Value Problem: Dirichlet Eigen value problem
2. Neumann Eigen value problem

- A **boundary value problem** is a differential equation where the initial conditions/conditions on the solution (or) its derivatives is specified at multiple points in the interval. On the contrary, the initial value problem (that was considered so far in this course) differential equation where the initial conditions/conditions on the solution and/or its derivatives is specified at one point in the domain.
- **Dirichlet Eigen value problem:** Consider a second-order boundary value problem

$$\frac{d^2y}{dx^2} + \lambda y = 0 \text{ with } y(0) = 0 \text{ and } y(L) = 0 \quad (1)$$

It is easy to validate that one trivial solution for (1) is $y \equiv 0$. However, the goal is to find λ for which a non-trivial solution exists for (1). The value/s of λ for which a non-trivial solution exists are called the **Eigen values** and the corresponding non-trivial solutions are called the **Eigen functions**¹. In order to determine the eigen values, consider the following 3 cases

- Case I ($\lambda = 0$): When $\lambda = 0$, the general solution of (1) is given by

$$y(x) = c_1 + c_2x \quad (2)$$

The boundary condition $y(0) = 0$, requires that $c_1 = 0$. With $c_1 = 0$, $y(x) = c_2x$. Further, $y(L) = 0$ requires that $c_2 = 0$. Thus, for $\lambda = 0$, the ODE (1) has a trivial solution $y(x) \equiv 0$.

- Case II ($\lambda = -\omega_n^2$): When $\lambda = -\omega_n^2$, the roots of the auxiliary equation are $s_1 = \omega_n$ and $s_2 = -\omega_n$ and the general solution of (1) is given by

$$y(x) = c_1 e^{\omega_n x} + c_2 e^{-\omega_n x} \quad (3)$$

The boundary condition $y(0) = 0$, requires that $c_1 + c_2 = 0$. With $c_2 = -c_1$, $y(x) = 2c_1 \sinh(\omega_n x)$. Further, $y(L) = 0$ requires that $c_1 \sinh(\omega_n L) = 0 \implies c_1 = 0$. Thus, for $\lambda = -\omega_n^2$, the ODE (1) has a trivial solution $y(x) \equiv 0$.

- Case III ($\lambda = \omega_n^2$): When $\lambda = \omega_n^2$, the roots of the auxiliary equation are $s_1 = +j\omega_n$ and $s_2 = -j\omega_n$ and the general solution of (1) is given by

$$y(x) = c_1 \cos(\omega_n x) + c_2 \sin(\omega_n x) \quad (4)$$

¹A more precise definition requires some background on PDEs

The boundary condition $y(0) = 0$, requires that $c_1 = 0$. With $c_1 = 0$, $y(x) = c_2 \sin(\omega_n x)$. Further, $y(L) = 0$ requires that $c_2 \sin(\omega_n L) = 0$. Thus, one solution of the ODE (1) when $\lambda = \omega_n^2$ is the trivial solution $y(x) \equiv 0$. However, in addition to the trivial solution, the boundary conditions, more precisely, the condition $y(L) = 0$, is satisfied when

$$\omega_n = \frac{n\pi}{L} \text{ where } n = 1, 2, \dots \quad (5)$$

In such a scenario, the solution is given by

$$y(x) = C \sin\left(\frac{n\pi}{L}x\right) \quad (6)$$

Thus, a nontrivial solution exists when

$$\lambda = \left(\frac{n\pi}{L}\right)^2 \text{ where } n = 1, 2, \dots \quad (7)$$

and the corresponding non-trivial solution (eigenfunctions) is given by

$$y(x) = C \sin\left(\frac{n\pi}{L}x\right) \text{ where } n = 1, 2, \dots \quad (8)$$

- **Periodic eigenvalue problem:** Consider a second-order boundary value problem

$$\frac{d^2 y}{dx^2} + \lambda y = 0 \text{ with } y(0) = y(L) \text{ and } y'(0) = y'(L) \quad (9)$$

It is easy to validate that one trivial solution for (9) is $y \equiv 0$. However, the goal is to find λ for which a non-trivial solution exists for (1). In order to determine the nontrivial solutions, consider the following 3 cases

- Case I ($\lambda = 0$): When $\lambda = 0$, the general solution of (1) is given by

$$y(x) = c_1 + c_2 x \quad (10)$$

The boundary condition $y(0) = y(L)$, requires that $c_2 = 0$. With $c_2 = 0$, $y(x) = c_1$. Further, for $y(x) = c_1$, $y'(L) = y'(0)$. Thus, for $\lambda = 0$, the ODE (9) has a nontrivial solution $y(x) = c$.

- Case II ($\lambda = -\omega_n^2$): When $\lambda = -\omega_n^2$, the roots of the auxiliary equation are $s_1 = \omega_n$ and $s_2 = -\omega_n$ and the general solution of (9) is given by

$$y(x) = c_1 e^{\omega_n x} + c_2 e^{-\omega_n x} \quad (11)$$

The boundary condition $y(0) = y(L)$, requires that

$$c_1 + c_2 = c_1 e^{\omega_n L} + c_2 e^{-\omega_n L} \quad (12)$$

and the boundary condition $y'(0) = y'(L)$, requires that

$$c_1 - c_2 = c_1 e^{\omega_n L} - c_2 e^{-\omega_n L} \quad (13)$$

Equations (13) and (13) hold only when $c_1 = c_2 = 0$ or $\omega_n = 0$, either of which results in trivial solution, $y(x) \equiv 0$.

- Case III ($\lambda = \omega_n^2$): When $\lambda = \omega_n^2$, the roots of the auxiliary equation are $s_1 = +j\omega_n$ and $s_2 = -j\omega_n$ and the general solution of (9) is given by

$$y(x) = c_1 \cos(\omega_n x) + c_2 \sin(\omega_n x) \quad (14)$$

The boundary condition $y(0) = y(L)$, requires that

$$c_1 = c_1 \cos(\omega_n L) + c_2 \sin(\omega_n L) \quad (15)$$

For non-zero c_1 and c_2 15 holds when $\omega_n L = 2\pi n$. Further, the boundary condition $y'(0) = y'(L)$, requires that

$$c_2 \omega_n = c_2 \omega_n \cos(\omega_n L) - c_1 \omega_n \sin(\omega_n L) \quad (16)$$

Equation (16) holds when $\omega_n L = 2\pi n$. Thus the nontrivial solution is given by

$$y(x) = c_1 \cos\left(\frac{2\pi n}{L}x\right) + c_2 \sin\left(\frac{2\pi n}{L}x\right) \quad (17)$$

for all c_1 and c_2 . Thus, the fundamental set of eigenfunctions for the eigenvalue $\lambda = \left(\frac{2\pi n}{L}\right)^2$ are $\cos\left(\frac{2\pi n}{L}x\right)$ and $\sin\left(\frac{2\pi n}{L}x\right)$.